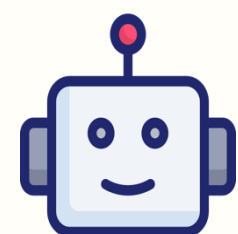




User Instruction

"Insert a blue sedan to the right of the white vehicle, slightly behind it, and make it go straight."

Orchestration Module



Orchestrator

```
def LangDriveCTRL(input_video, instruction):  
    scene_graph = decompose_scene(input_video)  
  
    # all functions can directly access the scene graph  
    # object query, locate the reference vehicle based on the text  
    white_vehicle = ground_object(obj_desc = "White Vehicle.")  
  
    # object node editing, generate and insert the blue sedan into the scene  
    blue_sedan = insert(obj_desc = "Blue sedan.")  
  
    # behavior editing, generate and refine the trajectory  
    # based on the feedback loop  
    gen_traj(obj = blue_sedan, behav_desc = "Go straight.",  
            init_pos_desc = "The right of the white vehicle, slightly behind it.")  
  
    # video rendering and iterative refinement  
    render_and_refine()
```

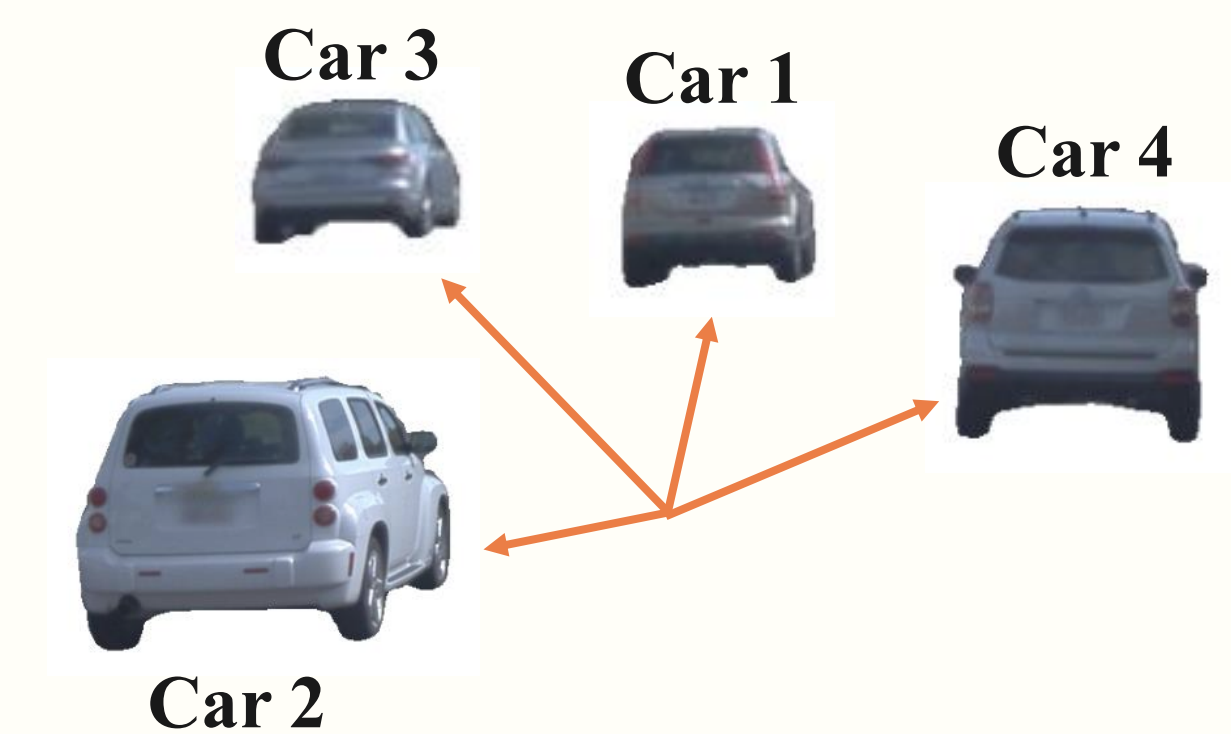
Input Video



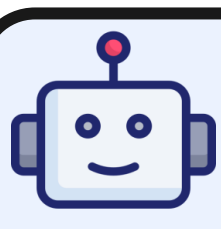
Scene Decomposition Module



3DGS Background Node



Dynamic Object Nodes
Scene Graph

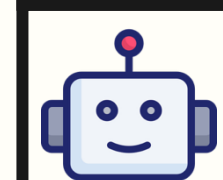


Agent



Tool

Object Query Module



"White vehicle."



Object Node Editing Module

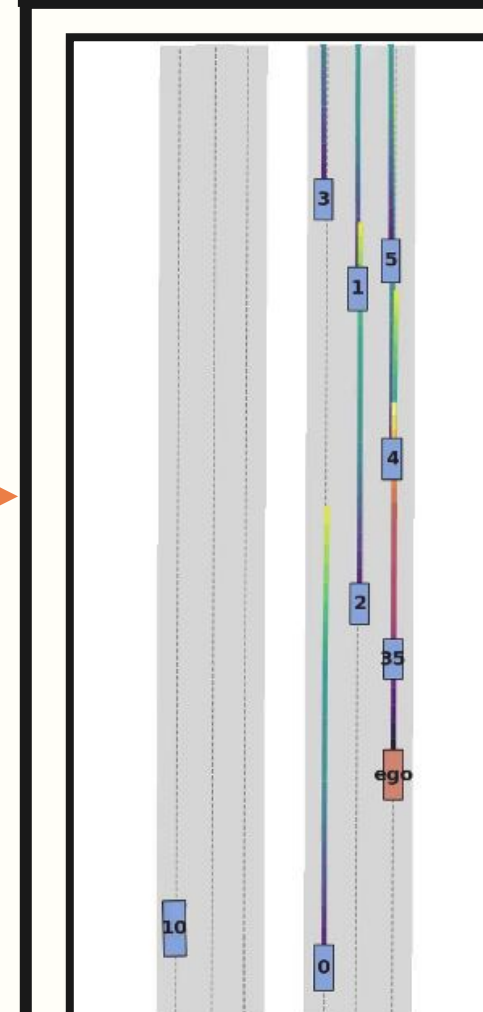


Text-to-3D Tool

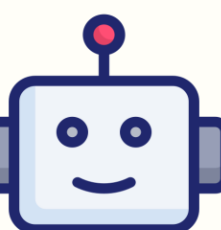
"Blue sedan."



Behavior Editing Module



Feedback
Loop



Behavior
Reviewer

Edited Video

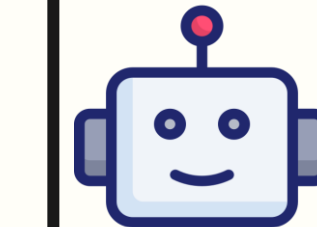


Rendering & Refinement Module



Video Diffusion
Tool

Feedback
Loop



Video
Reviewer



Refined Video



Coarse
Rendering Tool



Coarse Video